# Cloud Native Project - Cloud Application Development – Winter Term 2025/26

## Project team

# 1 Requirements

Give a short introduction into the system scope and main features.

## 1.1 System Context

Provide a **system context diagram** of the application and give a description of all neighboring systems, external interfaces and actors.

## 1.2 Feature Overview

Provide a brief description of the main features of the application.

## 1.3 Domain Model

Describe the main business entities, their relationships and main key attributes,

# 2 Runtime View

## 2.1 Runtime Overview

Describe the cloud resources (diagram) and the external interfaces and UI interfaces here. Describe the configurations of cloud resources used (e.g. API Gateway).

Describe how synchronous services (e.g. Web Frontends) and asynchronous services are implemented.

Provide links to the running application and the cloud environment.

## 2.2 Microservices

Give a detailed description of each microservice. Describe

- How is the service runtime configured?
- Is the microservice scalable? Is scaling handled manual or automatically?
- What is the security setup?
- What connections to external cloud services (e.g. datastores) exists?
- Highlight especially if/how isolation in the context of multi-tenancy is enforced.

## 2.3 Datastores

Which storage containers exist at runtime and give a link to the data model. Highlight especially if/how isolation in the context of multi-tenancy is enforced.

- Describe the different persistent storages your application uses
- For each storage, define the data model (e.g. ER-Diagram, JSON Schema, …)

## 2.4 Security: Roles and Role Mapping

Explain which service accounts exist and their privileges. Highlight especially if/how isolation in the context of multi-tenancy is enforced.

# 3 Development View

## 3.1 Software Components

Development time view on software components:

- Which repositories for the source code are used (link)? How is the repository organized?
- What are the software components of your application?
- In which programming languages is the software written? What frameworks are used? Which important libraries are used?

# 4 DevOps

## 4.1 Environments and Initial Infrastructure Setup

Starting from a blank cloud environment, what needs to be done to setup infrastructure before software can be deployed?

## 4.2 Pipelines and Release of new Features

Explain how software is built and deployed. Explain especially how the CI/CD pipeline builds artifacts and deploys them to test/production environments.  Explain your branching strategy.

## 4.3 Creation of new Tenants

Explain the process of tenant creation. Describe the automatic and the manual steps for each tenant type.

## 4.4 Monitoring

Explain how the health of the services is monitored and how and when alarms are triggered. Logging is a central part of Monitoring. Explain how logs are managed and how they can be queried.

# 5 Performance Tests

Analyze the performance of the application under various conditions.

## 5.1 Periodic Workload

Simulate the following scenarios

- 100 concurrent users during peak times and 10 during low demand times
- 1000 concurrent users during peak times and 20 during low demand times

Provide the following information:

- With which data is the application initialized before testing?
- Describe the transaction mix?
- Describe ramp-up/ramp-down phases of the load test, make sure you test long enough!
- Include report of response times, failure rates and resource utilization.

Analyze the results.

## 5.2 Once-in-a-lifetime Workload

Simulate the following scenarios

- Start with a base load of 10 concurrent users for a while and then add constantly new users.

Determine the max once-in-a-lifetime workload your application survives for a given growth rate of users.

- Which workload does the application survive without degradation?
- Which workload does the application survive with degradation?
- Which workload does the application no longer survive?

Provide the following information:

- With which data is the application initialized before testing?
- Describe the transaction mix?
- Describe ramp-up/ramp-down phases of the load test, make sure you test long enough!
- Include report of response times, failure rates and resource utilization.

Analyze the results.

# 6 Commerical Model

## 6.1 Tenant types

Give list of the tenant types and describe common functionality and where tenant types differ:

- Functional capabilities
- Thresholds
- Isolation

## 6.2 Pricing Model

Describe the pricing model for each tenant types. Define the parameters which determine the pricing and document free quotas and thresholds.

## 6.3 Cost Model

Estimate the operational costs for shared components and each tenant types and discuss best / avg / worst case scenarios.